HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and modified **PLC software** and **HMI applications** for control systems, integrating new automation components and troubleshooting existing issues to maintain operational performance.
- Produced and maintained comprehensive **technical documentation**, including functional specifications, software logic, and electrical schematics, ensuring clarity and adherence to engineering standards.
- Investigated and resolved issues with existing automated systems, implementing modifications and improvements to enhance reliability and technical performance across manufacturing equipment.
- Gathered and clarified **technical requirements** with cross-functional teams and stakeholders to define the scope and functionality of new automation projects.
- Provided **technical input** during the development, build, testing, and installation phases of automated equipment, ensuring alignment with project specifications and quality standards.
- Participated in system testing and **commissioning activities**, coordinating engineering efforts to meet agreed timelines and quality requirements for final handover.
- Containerized automation control applications using **Docker** and deployed to a managed cloud environment, configuring basic resource limits for predictable performance during updates.

SKILLS

Automation & Control – PLC Software Development, HMI Applications, Control Systems, Automation Components, System Testing, Commissioning

Software Development – Java Spring Boot, Python Scripting, REST APIs, Docker, Git, Jenkins

Data & Analytics – SQL (PostgreSQL, MySQL, Oracle), Power BI, Tableau, DAX, Power Query

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch)

Operations & Support – Production Support, Incident Response, SLA Management, Splunk, Dynatrace, Jira, Confluence

Engineering Documentation – Functional Specifications, Software Specifications, Hardware Specifications, Electrical Schematics, Engineering Drawings

PROJECTS

Manufacturing Process Automation Enhancement

- Developed new PLC logic and HMI screens for a critical manufacturing line, improving cycle time by **15%** and reducing manual intervention points.
- Integrated a new vision inspection system into the existing automation framework, requiring modification of control software and data logging to capture inspection results.
- Authored detailed technical documentation, including sequence of operations and hardware component lists, for the upgraded line, facilitating future maintenance and troubleshooting.

Automated Reporting and Data Integration Pipeline

- Built an automated Python ETL pipeline to extract data from multiple operational systems (PostgreSQL, Oracle) for compliance reporting, reducing manual data compilation effort by approximately **10 hours per week**.
- Developed a series of **Power BI dashboards** to visualize key operational metrics and system status, providing real-time insights to management and operations teams.
- Implemented robust error handling and logging within the Python scripts using **Flask**, ensuring data integrity and providing alerts for pipeline failures.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)